Bull Math Biol
DOI 10.1007/s11538-011-9676-y

ORIGINAL ARTICLE

A Continuum Three-Zone Model for Swarms

Jennifer M. Miller · Allison Kolpas · Joao Plinio Juchem Neto · Louis F. Rossi

Received: 24 October 2010 / Accepted: 21 June 2011
© Society for Mathematical Biology 2011

Abstract We present a progression of three distinct three-zone, continuum models for swarm behavior based on social interactions with neighbors in order to explain simple coherent structures in popular biological models of aggregations. In continuum models, individuals are replaced with density and velocity functions. Individual behavior is modeled with convolutions acting within three interaction zones corresponding to repulsion, orientation, and attraction, respectively. We begin with a variable-speed first-order model in which the velocity depends directly on the interactions. Next, we present a variable-speed second-order model. Finally, we present a constant-speed second-order model that is coordinated with popular individual-based models. For all three models, linear stability analysis shows that the growth or decay of perturbations in an infinite, uniform swarm depends on the strength of attraction relative to repulsion and orientation. We verify that the continuum models predict the behavior of a swarm of individuals by comparing the linear stability results with an individual-based model that uses the same social interaction kernels. In some unstable regimes, we observe that the uniform state will evolve toward a radially symmetric attractor with a variable density. In other unstable regimes, we observe an incoherent swarming state.

Keywords Swarms · Aggregation · Integrodifferential equation · Linear stability

J.M. Miller · A. Kolpas · J.P. Juchem Neto · L.F. Rossi (✉)
Department of Mathematical Sciences, University of Delaware, Newark, DE 19716, USA
e-mail: rossi@math.udel.edu

Present address:
A. Kolpas
Department of Mathematics, West Chester University of Pennsylvania, West Chester, PA 19383, USA
e-mail: akolpas@wcupa.edu

Published online: 29 July 2011

1 Introduction

Biological aggregations give members the benefit of protection, mate choice, and information that an individual might not be aware of on its own such as the location of a food source, predator, or migratory route (Camazine et al. 2003; Partridge 1982). There is considerable evidence in many species that individuals can only perceive their neighbors rather than the entire group (Camazine et al. 2003). In this case, groups are referred to as *self-organized* since they form without any centralized control. There are many examples of self-organized animal groups in nature including schools of fish, swarms of locusts, herds of wildebeest, and flocks of birds (Lissaman and Shollenberger 1970; Parrish 1999; Sinclair and Norton-Griffiths 1979; Uvarov 1928).

In self-organized groups, complex large-scale structures emerge through individual decisions based on input from local conditions. For example, in response to a predator, many schools of fish display complex collective patterns of motion, including compression, vacuole, flash expansion, and milling, or form highly parallel translating groups (Parrish et al. 2002). There have been successful efforts to map out the qualities of these structures as a function of the local interaction rules using individual-based modeling and computer simulation (Giardina 2008). In this paper, we explore the connection between local interaction rules and the emergent structures through mathematical analysis using a new continuum model.

In the Lagrangian approach, each organism is modeled as a discrete agent that interacts with other agents. The interactions are modeled as continuous-time or discrete-time dynamical systems and studied using techniques from nonlinear dynamics, control theory, or statistical mechanics (Cucker and Smale 2007; Huepe and Aldana 2004; Kolpas and Moehlis 2009; Kolpas et al. 2007; Levine et al. 2001; Paley et al. 2007b; Raymond and Evans 2006; Vicsek et al. 1995). Numerical simulations are frequently employed and are often the only means of analysis.

Individual-based models or agent-based models of swarming are discrete-time models which give the position and direction of travel of an individual at regular intervals of time as an explicit function of the positions and directions of other members of the group. Such models are capable of displaying a variety of different coherent structures with relatively simple interaction rules. One common approach is to place zones around individuals in which they respond to others through repulsion, alignment, and/or attraction (Aoki 1982; Couzin et al. 2002; Huth and Wissel 1992; Lukeman et al. 2010; Reuter and Breckling 1994; Vicsek et al. 1995; Warburton and Lazarus 1991).

In this paper, we will focus on the Couzin–Krause–James–Ruxton–Franks (CKJRF) model (Couzin et al. 2002) as a point of reference, but our methodology can be generalized to other swarming models based on zones. In the CKJRF model, interactions take place within three concentric zones around an individual: a "zone of repulsion," "zone of orientation," and "zone of attraction," the latter two excluding a blind volume behind the individual for which neighbors are undetectable. These zones are used to define behavioral rules of motion. First, if individual i finds agents within its zone of repulsion, it repels away from them by orienting its direction away from their average relative directions. Its desired direction of travel in the next time

step is given by

$$\mathbf{v}_i(t+\tau) = -\sum_{j\neq i} \frac{\mathbf{p}_j(t) - \mathbf{p}_i(t)}{|\mathbf{p}_j(t) - \mathbf{p}_i(t)|}, \tag{1}$$

and normalized as $\hat{\mathbf{v}}_i(t+\tau) = \frac{\mathbf{v}_i(t+\tau)}{|\mathbf{v}_i(t+\tau)|}$, assuming $\mathbf{v}_i(t+\tau) \neq 0$. If $\mathbf{v}_i(t+\tau) = 0$, individual i maintains its previous direction of travel giving $\hat{\mathbf{v}}_i(t+\tau) = \hat{\mathbf{v}}_i(t)$.

If agents are not found within individual i's zone of repulsion, then it aligns with (by averaging the directions of travel of itself and its neighbors) and feels an attraction toward (by orienting itself towards the average relative directions of) agents within the zone of orientation and attraction. Its desired direction of travel is given by the equally weighted sum of two terms:

$$\mathbf{v}_i(t+\tau) = \frac{\sum_{j\neq i} \frac{\mathbf{p}_j(t)-\mathbf{p}_i(t)}{|\mathbf{p}_j(t)-\mathbf{p}_i(t)|}}{|\sum_{j\neq i} \frac{\mathbf{p}_j(t)-\mathbf{p}_i(t)}{|\mathbf{p}_j(t)-\mathbf{p}_i(t)|}|} + \frac{\sum_j \hat{\mathbf{v}}_j(t)}{|\sum_j \hat{\mathbf{v}}_j(t)|}. \tag{2}$$

This vector is normalized assuming $\mathbf{v}_i(t+\tau) \neq 0$. If $\mathbf{v}_i(t+\tau) = 0$, then individual i maintains its previous direction of travel.

White noise is added to the desired direction vector in each case. The model is second-order: an individual turns a maximum of $\theta\tau$ radians toward its desired direction vector $\hat{\mathbf{v}}_i(t+\tau)$ and updates its position according to

$$\mathbf{p}_i(t+\tau) = \mathbf{p}_i(t) + s\hat{\mathbf{v}}_i(t+\tau)\tau, \tag{3}$$

where s is its (constant) speed of travel.

The three-zone model of Couzin et al. achieved milling, incoherent swarming, and translating steady-states by varying the relative ranges of interaction (Couzin et al. 2002). In the swarming state, the group is incoherent, and its center of mass remains approximately stationary in time. In the milling state, the group rotates and is locally aligned. Coherent translating groups are classified as "dynamically parallel" or "highly parallel" depending on their degree of alignment. Simulations of the model were limited to on the order of hundreds of individuals with the group arriving at a steady-state after an initial transient.

As Couzin et al. and others have demonstrated, simulation is a powerful tool for exploring the state space of possible swarm structures. To understand these structures, Lagrangian models remain the dominant means of investigating connections between individual-level behavior and population-level dynamics. However, it is difficult to mathematically analyze large groups using the Lagrangian formulation and theoretically predict parameter sensitivity and qualitative behavior of the entire aggregation. Eulerian models capture swarm dynamics by treating groups as a continuous field and modeling interactions as the evolution of local fluxes. Eulerian models rely upon the hypothesis that the motion of very large swarms can be described through an evolution equation for the density of individuals per unit volume. One advantage to this approach is that a mathematical model which consists of a system of partial differential equations does not change with the population size, whereas discrete models require a set of differential equations to govern the motion of every individual. Another advantage is that we can apply different analytical tools, often borrowed from

continuum mechanics rather than finite dimensional dynamical systems. In this paper, we develop and analyze *three* new Eulerian models in order to understand how the parameters that drive individual behavior affect the emergent behavior of the entire swarm.

There are several examples of the successful implementation of the continuum modeling approach. Mogilner and Edelstein-Keshet used attraction and repulsion in their integrodifferential equations and found conditions on the interaction terms that create a swarm with constant interior density and sharp edges (Mogilner and Edelstein-Keshet 1999). Topaz and Bertozzi used attraction, orientation, and repulsion in two dimensions (Topaz and Bertozzi 2004). In this model, the swarm takes on a vortex-like structure when attraction and repulsion are in balance. Topaz, Bertozzi, and Lewis developed a model with attraction and repulsion to search for stationary solutions with sharp edges rather than traveling solutions (Topaz et al. 2006). Eftimie et al. presented a one-dimensional continuum model that leads to a number of different patterns (Eftimie and de Vries 2007). This model incorporates the three social forces, and their numerical work indicated that all three forces were necessary.

We developed our new models to help to understand the structures observed in Couzin et al. (2002) and use the social interactions of repulsion, attraction, and orientation to determine the density and velocity field for a swarm. Each interaction takes place over a different region. We do not upscale the CKJRF model precisely for the following reasons:

- CKJRF uses eight different parameters. We choose to focus on the relative zone sizes and the relative influence of attraction.
- CKJRF is a discrete-time model. For simplicity, we use a continuous-time model.
- CKJRF uses discrete nonoverlapping zones. For analytical convenience, we choose to use continuous overlapping zones. We do not know of a biological determination of this issue at this level of detail.

Similar to other investigations, our process will result in partial integrodifferential equations to describe the motion of the swarm. Using linear stability analysis in the frequency (Fourier) domain, we will explore the linear stability of uniform infinite swarms. The results of the linear stability analysis of the continuum model are compared with corresponding individual-based simulations of finite swarms.

To connect our new model to CKJRF, we introduce three different models. First, we develop a first-order (or kinematic) model where the velocity field is a function of local density and velocities. This would map to CKJRF in the regime when the turning rate was much greater than the individual speed. Second, we develop a second-order (dynamic) model where acceleration is a function of local density and velocities. Finally, we develop a constant-speed, second-order model which maps closest to CKJRF. In each case, we will determine parameter regimes where uniform, constant-density, parallel groups are stable or not, and verify these regimes using individual-based simulations. In some cases where the uniform swarms are linearly unstable, we find that there is an attractor corresponding to a circular, translating swarm that has a variable density. We will also present individual-based simulations corresponding to the full nonlinear continuum model which includes a density dependent function that switches off orientation and attraction influences when the population density

reaches a critical threshold. The CKJRF model has a similar feature which causes an individual to respond based only on repulsion when any neighbors are present in the zone of repulsion. Simulations of the individual-based model in the regime for which uniform swarms are linearly unstable indicate the presence of an incoherent swarming state where the group's center of mass remains approximately fixed in time but individuals constantly adjust their velocity. This structure was observed in the CKJRF model.

This paper is organized as follows. This section surveys contributions to central issues about zonal swarm models. In Sect. 2, we will introduce our continuum model zones of repulsion, orientation and attraction. In Sect. 3, we describe our simulation methodology that will complement our analysis of the three-zone continuum models. In Sect. 4, we explore a simple first-order (kinematic) continuum model for three-zone swarming. In Sect. 5, we analyze and explore a second-order (dynamic) model that includes a simple, three-zone control process. In Sect. 6, we restrict this dynamic model to constant-speed solutions. This last model is directly aligned with popular individual-based models.

2 Three-Zone Continuum Models

When developing our continuum model, we assume that the individuals in a swarm are following some basic rules of thumb and that individuals make decisions based on only their neighbors' positions and velocities rather than using information about every member of the swarm. The rules we use are the same as those used in the CKJRF model:

- Individuals will attempt to maintain a minimum distance between themselves and others. This response will be the only response if the density is high.
- If the density is not too high, then in addition to repulsion individuals orient with the average direction of neighbors and are attracted to other individuals at a distance.

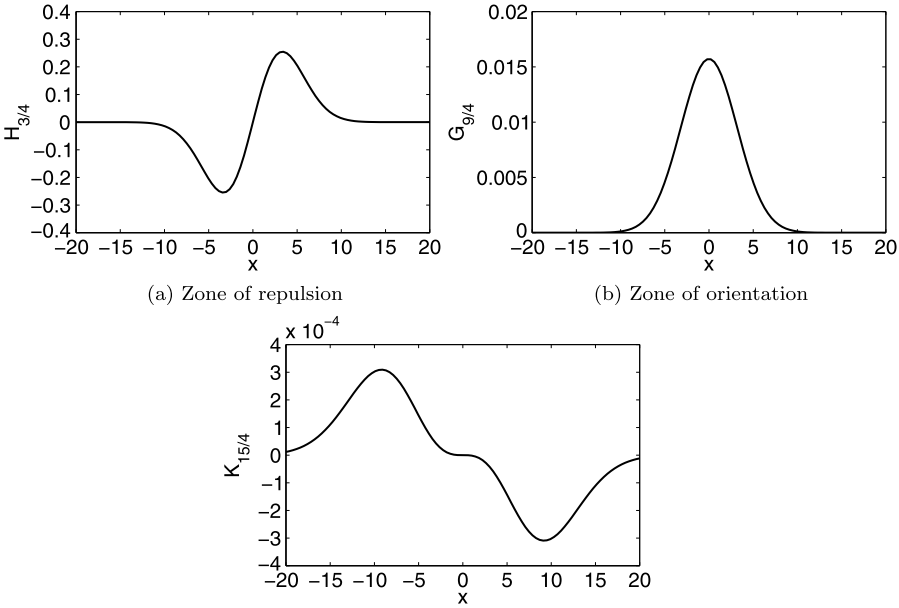
All individuals follow the same rules and no external influences are added. Influences from the zone of repulsion keep individuals from colliding. Attraction helps the group maintain cohesion and avoid fragmenting. Influences from the zone of orientation may lead the group to arrive at a consensus around a common alignment or other coherent structures such as vortices or milling. In CKJRF, the key parameter is the size of the zones relative to one another.

We use the same social interaction kernels in all three models.

$$H_{\sigma_1} = \frac{1}{8\pi\sigma_1^4}\mathbf{x}e^{-|\mathbf{x}|^2/4\sigma_1^2} \quad \text{(repulsion)}, \tag{4a}$$

$$G_{\sigma_2} = \frac{1}{4\pi\sigma_2^2}e^{-|\mathbf{x}|^2/4\sigma_2^2} \quad \text{(orientation)}, \tag{4b}$$

$$K_{\sigma_3} = \frac{-1}{64\pi\sigma_3^6}\mathbf{x}|\mathbf{x}|^2e^{-|\mathbf{x}|^2/4\sigma_3^2} \quad \text{(attraction)}. \tag{4c}$$



(a) Zone of repulsion

(b) Zone of orientation

(c) Zone of attraction

Fig. 1 Cross-sections of continuous overlapping kernels with $\sigma_1 = 3/4$, $\sigma_2 = 9/4$, and $\sigma_3 = 15/4$

The widths of the zones of repulsion, orientation, and attraction are σ_1, σ_2, and σ_3, respectively. There are several quantitative differences between the zones used in the CKJRF model and our continuous model. For analytical convenience, we chose to use functions with Gaussian decay. Even though they do not have compact support, these kernels guarantee that any influence from distant individuals will be exponentially small relative to neighbors. Furthermore, the CKJRF zones are disjoint whereas the continuous zones overlap. The continuous overlapping zones lead to smooth transitions in individual behavior rather than abrupt switching that can occur in the original CKJRF model. Sample plots of the continuous, overlapping zones are shown in Fig. 1.

For convenience, we restrict our examples to two spatial dimensions and normalize our kernels assuming $d = 2$. We normalize our kernels so that for the test function $\phi(\mathbf{x}) = \mathbf{a} \cdot \mathbf{x} + b$,

$$H_{\sigma_1} \star \phi = -\mathbf{a}, \qquad K_{\sigma_3} \star \phi = \mathbf{a}, \tag{5}$$

and for $\psi(\mathbf{x}) = b$,

$$G_{\sigma_2} \star \psi = b. \tag{6}$$

Our goal is to create a continuum model that predicts the same global behavior as an individual-based model, in particular the constant-speed second-order discrete-time model presented in Couzin et al. (2002). To achieve this goal, we systematically develop three models.

1. A first-order (kinematic) model. This is the simplest three-zone model where the velocity is a function of the densities and velocities in the three zones.
2. A second-order (dynamic) model. This is a natural progression of the first model to include a simple control term. The desired velocity is a function of the densities and positions in the three zones. A simple control function moves the current velocity vector toward the desired velocity vector.
3. A second-order, constant-speed model. This is an extension of the second-order model that restricts the evolution of the velocity vector to rotation only. With this constraint, this continuum model is closest to the CKJRF swarming model.

In all three models, a uniform, translating swarm ($\rho =$ constant, $\mathbf{v} =$ constant) is a stationary state of the system. We interpret this solution to be the "highly parallel group" which occupies a large fraction of the state space studied computationally in Couzin et al. (2002). One of the key discoveries in Couzin et al. (2002) was that "parallel groups" emerge and are robust to perturbations for certain zone sizes. Of course, in individual-based simulations, the swarms are of finite extent, but results for infinite swarms are valid for regions of finite swarms where the significant portions of the interaction kernels do not extend beyond the boundary. Therefore, the stability of uniform infinite swarms in these models can inform us about the fate and evolution of large finite swarms.

3 Individual-Based Simulations

Before exploring the continuum three-zone models, we will discretize the social interaction terms so that they can be directly compared with interaction terms of the CKJRF model. Ideally, the discretization would be exactly the same, but there are a number of reasons why this is not the case. Particularly, normalizing the mean of unit vectors in (1) and (2) can lead to spurious results. For instance, in a near-constant density swarm, infinitesimal variations will lead to $O(1)$ accelerations. Our formulation avoids these difficulties.

To discretize the continuum model, we replace the density with a collection of N individuals with positions $\mathbf{x}_i$ and velocities $\mathbf{v}_i$ for $i = 1, \ldots, N$. Individuals synchronously update their positions and velocities at regular time intervals of length τ, making movement decisions based on the behavioral input of neighbors according to the social interaction kernels.

Consider a reference individual i and define the displacement vector $\mathbf{x}_{ij} = \mathbf{x}_j - \mathbf{x}_i$. We define a behavioral input vector $\mathbf{v}_{d,i}$ as a weighted sum of the velocity vector contributions from repulsion, orientation, and attraction zones

$$\mathbf{v}_{d,i} = \mathbf{v}_{r,i} + f(\rho)[\mathbf{v}_{o,i} + c_a \mathbf{v}_{a,i}], \quad i = 1, \ldots, N \tag{7}$$

where

$$\mathbf{v}_{r,i} = \sum_{j=1}^{N} -\frac{1}{8\pi\sigma_1^4}\mathbf{x}_{ij} \exp\left(-|\mathbf{x}_{ij}|^2/4\sigma_1^2\right), \tag{8a}$$

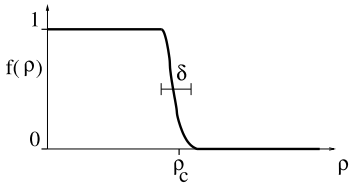


Fig. 2 Graph of switching function $f(\rho)$. The parameter δ is assumed to very small

$$\mathbf{v}_{a,i} = \sum_{j=1}^{N} \frac{1}{64\pi\sigma_3^6} \mathbf{x}_{ij} |\mathbf{x}_{ij}|^2 \exp\left(-|\mathbf{x}_{ij}|^2/4\sigma_3^2\right), \tag{8b}$$

$$\mathbf{v}_{o,i} = \frac{\sum_{j=1}^{N} \frac{1}{4\pi\sigma_2^2} \exp(-|\mathbf{x}_{ij}|^2/4\sigma_2^2)\mathbf{v}_j}{\sum_{j=1}^{N} \frac{1}{4\pi\sigma_2^2} \exp(-|\mathbf{x}_{ij}|^2/4\sigma_2^2)}. \tag{8c}$$

The parameter c_a weights the strength of attraction relative to the other two behavioral interaction terms. Similar behavioral interaction terms appear in the three-zone individual-based model of Couzin et al. (2002) but with all individuals (within a zone) weighted equally. The switching function $f(\rho)$ has the property that $f(\rho) = 1$ when $\rho < \rho_c - \delta/2$ and $f(\rho) = 0$ when $\rho > \rho_c + \delta/2$ (see Fig. 2). The purpose of this function is to connect the continuum model with the CKJRF model in which only repulsion acts when an individual is within the zone of repulsion. One interpretation is that orientation and attractive forces are excluded when the density exceeds a critical threshold ρ_c.

The contribution from repulsion (attraction) results in individuals moving away (towards) a weighted average of the relative directions of all other swarm members. We can compare (1) and (8a) directly if we associate $\mathbf{p}_j - \mathbf{p}_i$ with $\mathbf{x}_{ij}$. The CKJRF sums the individuals equally while we apply an exponential envelope. The CKJRF model normalizes individuals, takes the mean, and then normalizes again which can lead to spurious switching even in situations with a near-constant density, constant velocity swarm. With normalization properties (5) in mind, the discretization of the continuum model sums and then normalizes and avoids spurious switching. The same connections can be drawn between the attraction term on the left side of (2) and (8b). The contribution from the orientation zone is a weighted average of the velocities of all members of the swarm including the reference individual. Similarly, we compare the right side of (2) directly with (8c) along with the normalization (6). In the CKJRF model, unit vectors from the constant speed model are summed and then normalized. In the discretization of the continuum model, we sum the velocity vectors and then normalize the result from (6). The only difference is the exponential envelope.

The behavioral input vector $\mathbf{v}_{d,i}$ will be used in different ways in each of our three proposed models to determine an individual's velocity. See Sects. 4, 5, and 6 for more details. Once the velocities of each individual swarm member are updated, positions are integrated forward in time according to a forward (explicit) Euler scheme with step size τ:

$$\mathbf{x}_i(t+\tau) = \mathbf{x}_i(t) + \tau \mathbf{v}_i(t), \quad i = 1, \ldots, N. \tag{9}$$

Such a numerical integration scheme is typical in individual-based models for swarming where τ is interpreted as an individual's response time. We can also interpret τ as a numerical parameter controlling the integration of the differential equation

$$\frac{d}{dt}\mathbf{x}_i = \mathbf{v}_i. \tag{10}$$

Typically, τ is chosen to be small relative to the time scales over which the system changes so that the individual-based simulations are well resolved with this interpretation. We solved the individual-based model using the high-order adaptive `odeint` solver in SciPy based on ODEpack to verify that all solutions presented in this paper were fully resolved.

4 The First-Order (Kinematic) Model

All three models must conserve mass:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \tag{11}$$

where ρ is the density and $\mathbf{v}$ is the velocity field of the swarm. However, the individual behavior is represented by an additional equation describing the evolution of $\mathbf{v}$. What distinguishes each of our three models is the behavioral term.

In the first-order model, we assume that individuals respond instantly to social cues from the three zones:

$$\mathbf{v} = H_{\sigma_1} \star \rho + f(\rho)\big(G_{\sigma_2} \star (\rho \mathbf{v})(G_{\sigma_2} \star \rho)^{-1} + c_a K_{\sigma_3} \star \rho\big). \tag{12}$$

Individual speed and orientation can vary in this model. Since the response is instantaneous, the behavioral input vector is the velocity. In (12), the term $H_{\sigma_1} \star \rho$ contributes the repulsion effect. Orientation comes from the term $G_{\sigma_2} \star (\rho \mathbf{v})(G_{\sigma_2} \star \rho)^{-1}$, and attraction is caused by the term $K_{\sigma_3} \star \rho$. As in the individual-based simulations, the function $f(\rho)$ acts as a behavioral switch which turns off orientation and attraction when the population reaches a critical density and the parameter c_a controls the importance of attraction as compared to orientation and repulsion.

Another distinctive feature of the first-order model is that solutions to (12) are not unique. For example, for an infinite swarm with a constant density, attractive and repulsive terms will be zero. Assuming that $f(\rho) = 1$ for this constant density, then

$$\mathbf{v} = G_{\sigma_2} \star (\rho \mathbf{v})(G_{\sigma_2} \star \rho)^{-1} = G_{\sigma_2} \star \mathbf{v}. \tag{13}$$

A constant velocity field in any direction will satisfy this expression. Biologically, this is not unreasonable. An analogy could be drawn to a group of individuals committing to travel together and remain together. This commitment requires individuals to align with one another, but does not determine a specific direction. In this case, coordination rules place some restrictions on the swarm configuration, but additional information from other sources, such as memory or internal preferences, must be supplied to arrive at a unique decision.

An infinite swarm with constant density ρ_0 and velocity $\mathbf{v}_0$ is the simplest equilibrium solution. We want to understand the effect of a small disturbance in the density, so we linearize the equations for the model and add a plane wave perturbation $\rho^{(1)} = Ae^{i(\xi\cdot\mathbf{x}-\omega t)}$ to the density. We then can examine the resulting disturbance $\mathbf{v}^{(1)}$ in the velocity.

We assume that the constant density ρ_0 is not high enough to make repulsion a priority, so $f(\rho_0) = 1$ and $f'(\rho_0) = 0$. Then our equation for $\mathbf{v}^{(1)}$ is

$$\mathbf{v}^{(1)} = H_{\sigma_1} \star \rho^{(1)} + G_{\sigma_2} \star \mathbf{v}^{(1)} + c_a K_{\sigma_3} \star \rho^{(1)}. \tag{14}$$

Taking the Fourier transform defined by $\widehat{g}(\xi) = \int_{-\infty}^{\infty} g(x)e^{i\xi x}\,dx$ and combining the two equations allows us to obtain the dispersion relation

$$\omega = -\mathbf{v}_0 \cdot \xi - i\rho_0|\xi|^2 \frac{e^{-\sigma_1^2|\xi|^2} + c_a(\frac{\sigma_3^2}{2}|\xi|^2 - 1)e^{-\sigma_3^2|\xi|^2}}{1 - e^{-\sigma_2^2|\xi|^2}}. \tag{15}$$

See Appendix A for full details of the linear stability analysis.

Perturbations in the swarm will grow when the imaginary part of ω is positive, which is equivalent to

$$e^{-\sigma_1^2|\xi|^2} + c_a\left(\frac{\sigma_3^2}{2}|\xi|^2 - 1\right)e^{-\sigma_3^2|\xi|^2} < 0. \tag{16}$$

We assume that $\sigma_1 < \sigma_3$ so that repulsion interactions are significant in an area close to the origin and attraction is important in an area further away. In order to determine when (16) holds, we make the change of variables $x = \sigma_3^2|\xi|^2$ and $\alpha = (\sigma_1/\sigma_3)^2$ so that we can consider $e^{-\alpha x} + c_a(\frac{x}{2} - 1)e^{-x} < 0$ or equivalently

$$e^{x(1-\alpha)} + c_a\left(\frac{x}{2} - 1\right) < 0. \tag{17}$$

If we define $h(x) = e^{x(1-\alpha)} + c_a(\frac{x}{2} - 1)$ for $x \geq 0$, then h is increasing and so the minimum of h occurs at $x = 0$. If $0 < c_a \leq 1$, then $h(x)$ is nonnegative for all $x > 0$. In this case, the imaginary part of ω will always be nonnegative and a plane wave perturbation with any wavenumber will not grow. If $c_a > 1$, then $h(0) < 0$ and since h is continuous, there is an interval around $x = 0$ where $h(x)$ is negative. Wavenumbers such that $x = \sigma_3^2|\xi|^2$ falls into this interval will make $\Im(\omega) > 0$, and so plane wave perturbations with these wavenumbers will grow exponentially. In other words, disturbances to the uniform state will die out if $c_a > 1$ and the perturbation has a sufficiently large wavenumber or if $c_a \leq 1$.

We first verified this result using particle-based simulations by discretizing (12) in a $[0, L]^2$ periodic box to see if the longwave instabilities would grow from a small seeded disturbance. To approximate a constant density of $\rho = 1$, we place individuals at positions $\mathbf{x}_k$ on an $L \times L$ regular grid. Following the discretization procedure described in Sect. 3 with $N = L^2$, we must solve a linear system $Av = b$ to determine the individual velocities:

$$A_{ij} = \delta_{ij} - \frac{G_{\sigma_2}(\mathbf{x}_i - \mathbf{x}_j)}{\sum_{j=1}^{N} G_{\sigma_2}(\mathbf{x}_i - \mathbf{x}_j)}, \tag{18a}$$

$$b_i = \sum_{j=1}^{N} \big[H_{\sigma_1}(\mathbf{x}_i - \mathbf{x}_j) + c_a K_{\sigma_3}(\mathbf{x}_i - \mathbf{x}_j) \big], \tag{18b}$$

where δ_{ij} is the Kronecker delta, and it is understood that v and b have a component for each direction. For instance, when $d = 2$, we must solve two linear systems. Notice that (18a) represents interactions according to (8c). As noted earlier, this linear system has a nontrivial null space.

This experiment is limited by the fact that the longest possible wavelength is the box size L, so that numerically, the smallest nontrivial wavenumber is $2\pi/L$. We use $\sigma_1 = 3/4$, $\sigma_2 = 9/4$, $\sigma_3 = 15/4$, and $L = 30$ for these experiments. With this configuration, we plot $\Im(\omega)$ for the first 20 discrete ξ's in Fig. 3. The disturbance $\xi_0 = 0$ is a translation mode and can be disregarded. Notice that the $\Im(\omega)$ is nearly zero at ξ_1 even when $c_a = 3$ because the function turns around so quickly near $\xi = 0$. For $c_a = 10$, we can see strong growth for ξ_1. We verify this instability by using Euler's method to integrate (11) using (18) to determine the velocity. Since the linear system has a nontrivial null space, we use the Moore–Penrose pseudo-inverse (Penrose and Todd 1956) as our particular solution, $v_p(\mathbf{x}_k)$. This procedure will produce the particular velocity field with minimum norm. To this velocity vector, we can add a null vector from (18). For instance, if we desired a swarm that translates horizontally, we could add a null vector to the $\hat{x}$ component of $\mathbf{v}_p$. We use a time step size of 0.1 which is adequate for resolving the instability, and march forward until $T = 2$. For initial conditions, we seeded the regular array by adding $10^{-2}[\sin(2\pi x/L) + \sin(4\pi x/L)]$ to each initial individual position. In Fig. 4, we can see the initial and final positions of the individuals, demonstrating the growth of the unstable mode, ξ_1, but not the second mode, ξ_2, as predicted. If we reduce c_a below 3, the instability disappears for this discretization as predicted by the analysis.

This result is consistent with the fact that extremely large swarms are not often observed in nature and that larger swarms are more likely to fragment than smaller ones. Indeed, recent research suggests that animal group-size distributions follow a power-law decay, favoring small groups over extremely large ones (Bonabeau and Dagorn 1995; Bonabeau 1999; Niwa 2003). One can interpret orientation interactions as a drive toward consensus while attractive interactions are a drive toward cohesion.

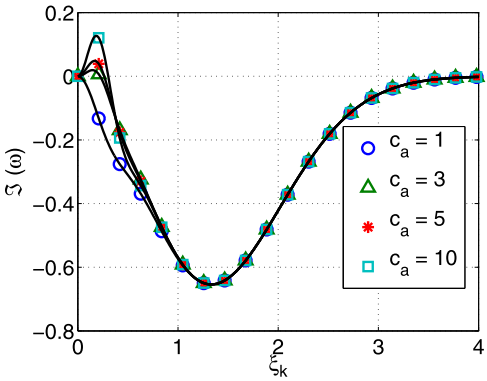


Fig. 3 Growth rate of discrete modes for a discretized square box of width $L = 30$, so that $\xi_k = 2\pi k/L$ where $k = 0, 1, 2, \ldots$. Modes will grow when $\Im(\omega) > 0$ (see (15))

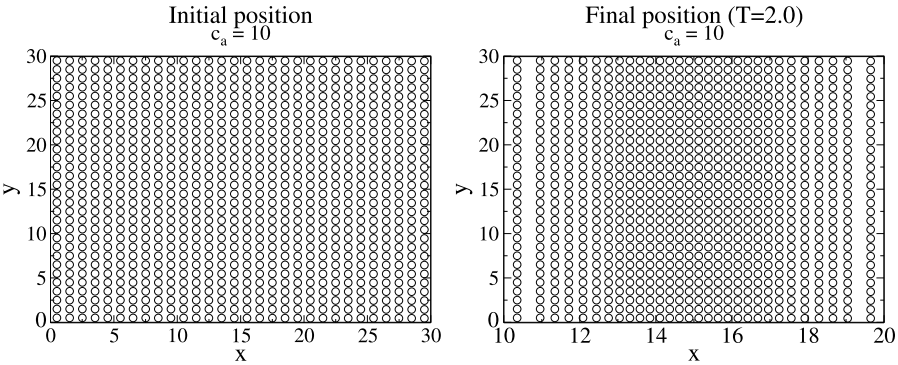


Fig. 4 Verification of linear stability results. The predicted unstable mode grows for $c_a = 10$ with $\sigma_1 = 3/4$, $\sigma_2 = 9/4$, $\sigma_3 = 15/4$ and $L = 30$

Our parameter c_a controls the relative importance of one over the other. When cohesion becomes excessive relative to consensus, the constant density translating state destabilizes. Mathematically, the largest possible wavelength that can be expressed is the size of the swarm itself which limits the smallest available wavenumber. A zero wavenumber disturbance is associated with a uniform increase or decrease in density without any spatially varying perturbation.

To place the result from the linear stability analysis in the context of the individual-based model, we consider a large finite swarm on a regularly spaced lattice. Individuals are initialized with velocity vector $\langle 1, 0\rangle$ so that the swarm translates horizontally. In contrast to the particle-based simulations, the individual-based model simulations are performed iteratively, and we do not impose periodic boundary conditions.

At each time step, the linear system (18) must be solved to determine the new velocities of the members of the swarm before updating their positions according to an explicit Euler scheme. In terms of the behavioral input vector $\mathbf{v}_{d,i}$ as defined in Sect. 3, this is equivalent to solving the implicitly defined linear system

$$\mathbf{v}_i = \mathbf{v}_{d,i}, \quad i = 1, \ldots, N. \tag{19}$$

Note that $\mathbf{v}_{d,i}$ is a function of $\mathbf{v}_j$ and $\mathbf{x}_{ij}$ for $j = 1, \ldots, N$.

We use a Jacobi fixed-point iteration to solve (19) iteratively for the velocity vectors of the swarm members according to

$$\mathbf{v}_i^{(k+1)} = \mathbf{v}_{d,i}^{(k)}, \quad i = 1, \ldots, N, \; k = 0, 1, 2, \ldots. \tag{20}$$

The algorithm is seeded at step $k = 0$ with the current positions and velocities of all swarm members from the previous Euler step. The velocities are then iterated while leaving the positions fixed until the swarm has reached a consensus (the norm of the residual of each component is less than $\mathrm{TOL} = 5 \times 10^{-3}$). Once the velocities are determined, positions are integrated forwards in time as described in Sect. 3.

The individual-based simulations are for finite swarms, but we expect the linear stability results to be valid in regions where the zones do not extend beyond the

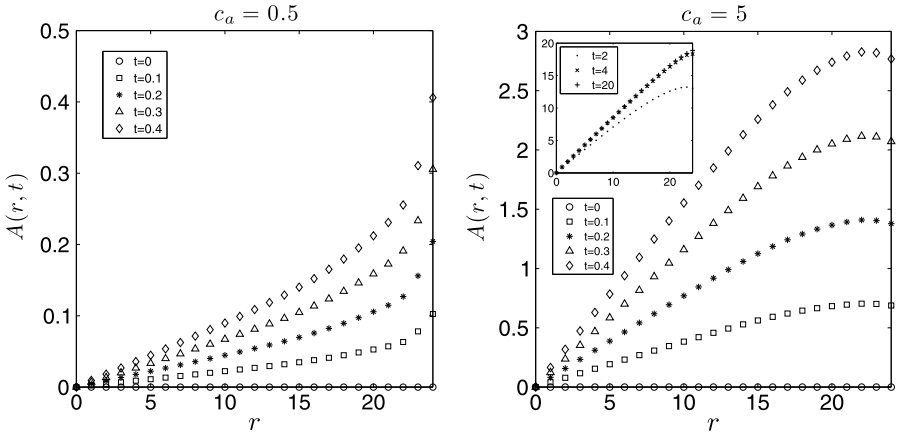


Fig. 5 Comparison of the first-order individual-based model dynamics when $c_a = 0.5$ (stable) and $c_a = 5$ (unstable). The results are for a swarm of size $N = 2401$ with $\sigma_1 = 3/4$, $\sigma_2 = 9/4$, and $\sigma_3 = 15/4$. A coarse-observable $A(r,t)$ is used to measure the magnitude of change in configuration over time as a function of distance from the interior

boundary. To characterize the swarm dynamics and verify stability, we define the following coarse observable:

$$A(r,t) = \begin{cases} \sum_{\|\mathbf{x}_i(0)\|_\infty = r} \|\mathbf{x}_i(0) + t\langle 1,0\rangle - \mathbf{x}_i(t)\|, & \text{for } r = 0, \\ \frac{1}{8r} \sum_{\|\mathbf{x}_i(0)\|_\infty = r} \|\mathbf{x}_i(0) + t\langle 1,0\rangle - \mathbf{x}_i(t)\|, & \text{for } r > 0. \end{cases} \tag{21}$$

The observable $A(r,t)$ gives the average over square rings of the positional displacement (using the Euclidean norm) of individuals from time step 0 to time step t in a moving reference frame. For a regular square lattice of length $L = 49$, $r = 0, 1, \ldots, 24$, where $r = 0$ corresponds to the individual at the center of the swarm and $r = 24$ are the individuals along the boundary. Thus, $A(r,t)$ is a measure of the magnitude of change in the swarm's configuration over time as a function of distance from the interior.

As noted for the simulations on a periodic lattice, our linear stability experiments are limited by the swarm size. We use the same kernel parameters but simulate a larger swarm of size $N = 49^2 = 2401$. The computation of $\Im(\omega)$ for the first few discrete ξ's for this size swarm indicates that we should not expect to see any instability below approximately $c_a = 2$ or substantial growth of unstable modes until approximately $c_a = 5$.

Figures 5 and 6 show the results of the individual-based model simulations with $c_a = 0.5$ and $c_a = 5$. Since this is a first-order (kinematic) model, all swarm members instantaneously interact with each other. Thus, boundary effects are felt throughout the swarm even over short timescales. This is in contrast to the second-order variable-speed model for which individuals in the interior of the swarm do not feel any substantial boundary effects over short timescales. See Sect. 5 for more details. Even so, there is a distinct difference in the dynamics of the individual-based model when $c_a = 0.5$ (linearly stable) and $c_a = 5$ (linearly unstable). When $c_a = 0.5$, $A(r,t)$ is a

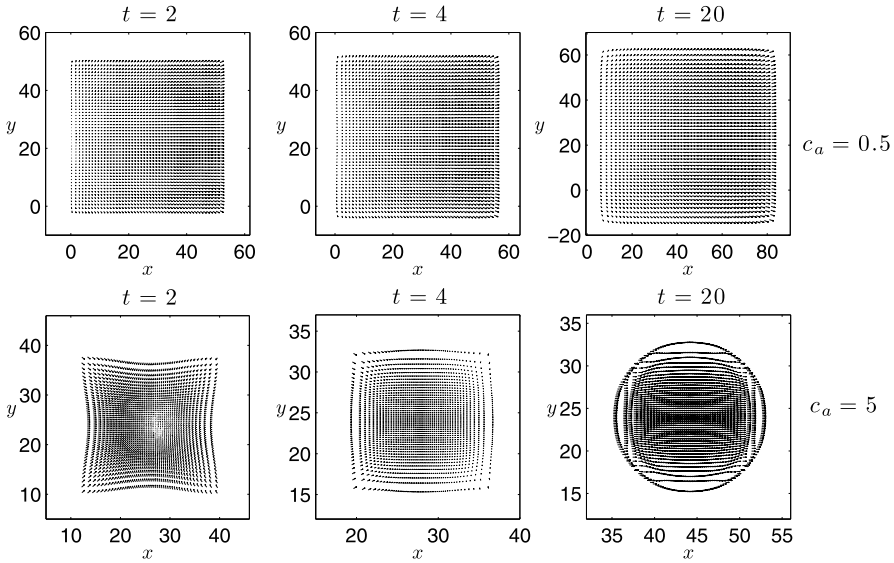


Fig. 6 Snapshots at times $t = 2, 4, 20$ of swarms simulated using the individual-based model with $N = 2401$, $\sigma_1 = 3/4$, $\sigma_2 = 9/4$, $\sigma_3 = 15/4$. When $c_a = 0.5$ (*top*), the uniform translating state expands at a relatively constant rate due to boundary effects. When $c_a = 5$ (*bottom*), the swarm evolves away from the uniform translating state toward a circular variable-density attractor

monotonically increasing function of r. This makes sense since boundary effects play a greater role the further you move from the interior of the swarm. In contrast, when $c_a = 5$, $A(r, t)$ is not simply monotonic but is concave, initially increasing and then decreasing as it approaches the boundary. Such spatial variation in the magnitude of change in configuration at the boundary suggests an instability. Indeed, long-time simulations reveal that in the unstable parameter regime, the swarm evolves away from the uniform translating state toward an axisymmetric, variable-density, coherent attractor. The observable $A(r, t)$ captures this, settling down to a steady function of r for sufficiently large t as shown in the inset of Fig. 5. If we reduce c_a below 2, the instability disappears for this discretization as predicted by our analysis.

Motivated by the result shown in Fig. 6 for $c_a = 5$, we seek an axisymmetric coherent structure. If we assume a constant velocity field in (12) with $f(\rho) = 1$, we get the following equation for the steady-state:

$$(H_{\sigma_1} + c_a K_{\sigma_3}) \star \rho = \mathbf{0}, \tag{22}$$

which can be written in the form:

$$\int_{\mathbb{R}^2} \left[H_{\sigma_1}(\mathbf{x} - \boldsymbol{\xi}) + c_a K_{\sigma_3}(\mathbf{x} - \boldsymbol{\xi}) \right] \rho(\boldsymbol{\xi})\, d\boldsymbol{\xi} = \mathbf{0}. \tag{23}$$

Of course, this homogeneous integral equation admits the trivial solution $\rho \equiv 0$, but since the integral operator $H_{\sigma_1} + c_a K_{\sigma_3}$ has a nontrivial null space, it also admits nontrivial solutions. We seek nontrivial, axisymmetric solutions $\rho \equiv \rho(r)$ to (23) that

also satisfy reasonable auxiliary constraints. For instance, (23) does not require any sort of regularity, but we seek solutions that are regular on the scales larger than the typical separation between individuals in the swarm. Just as importantly, we seek solutions such that $\rho(r \geq R) = 0$. Finally, (23) is a homogeneous equation, so we impose constraints to select a specific solution. We will specify the total mass of the solution $\iint \rho \, dA$. Applying all these constraints, we arrive at the following minimization problem:

$$Q = \min_{\rho} \left\{ \left\| (H_{\sigma_1} + c_a K_{\sigma_3}) \star \rho \right\|_2 + \lambda_1 \|\nabla \rho\|_2 + \lambda_2 \big(\rho(R)\big)^2 + \lambda_3 \left(\iint_D \rho \, dA - \bar{m} \right)^2 \right\} \tag{24}$$

where λ_1, λ_2, λ_3 are positive penalty parameters, and $\|\cdot\|_2$ is the L^2 norm, defined by $\|g\|_2 := (\iint_D g^2(x)\, dA)^{1/2}$ and D is the disk of radius R, centered at the origin. $\bar{m} = \pi R^2$ is the total mass of individuals in the disk.

To approximate ρ, we discretize the domain uniformly in the radial direction and minimize Q in (24) on this domain using SciPy's implementation (`fmin_bfgs`) of the Broyden–Fletcher–Goldfarb–Shanno (BFGS) optimization method (see Bonnans et al. 2006 for a general reference). The penalty terms with λ_1, λ_2 and λ_3 correspond to the variation of ρ on the domain, the density on the boundary, and the total mass of the system, respectively. To select the minimum variation solution with the appropriate constraints, we minimize the functional for small penalty values (λ_k's), and then gradually raise the penalties in discrete steps using previous solutions as the initial guess. The ratio of λ_1/λ_2 and λ_1/λ_3 is kept fixed as we raise λ_1. The converged solution is not sensitive to this ratio.

In Fig. 7, we plot the results of three minimized solutions of (24) where r is discretized over $N_R = 100$ points with the same parameters as in Fig. 6 ($c_a = 5$). The three solutions demonstrate the convergence of the optimization regime as the penalties grow. The solid curve with $\lambda_1 = 2.5$ and $\lambda_2 = \lambda_3 = 1000$ is indistinguishable from solution curves with larger penalties. Similarly, converged solutions using larger values of $N = 200$ and 400 are indistinguishable from the solution shown in the figure. To measure ρ from the individual-based model, we divide the computational domain into annuli of equal area and count the number of individuals in each annulus. To explore how close our individual-based model is to the continuum formulation, we scale the swarm up by a factor of 1.5, keeping the density the same, so that $N = 5476$, $\sigma_1 = 9/8$, $\sigma_2 = 27/8$, $\sigma_3 = 45/8$, $c_a = 5$. In Fig. 7, we see that the theoretical curve represents the data well with the exception of one outlier which is likely an artifact of variations when N is small and there are small numbers of individuals in each annuli.

5 The Second-Order (Dynamic) Model

The second-order variable-speed model is close to the first-order model but incorporates a simple control process. The behavioral input vector field $\mathbf{v}_d$ depends on the

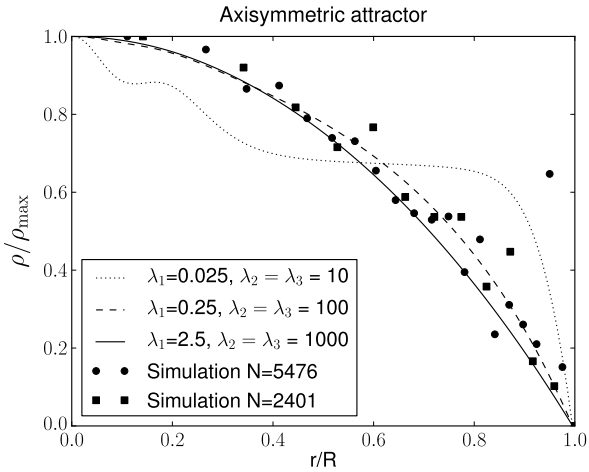


Fig. 7 Density profile of the axisymmetric attractor for $c_a = 5$: continuum minimizing solution versus individual-based model simulations. We compare density profiles of the axisymmetric attractor at regular resolution (Fig. 5) and high resolution ($N = 5476$, $\sigma_1 = 9/8$, $\sigma_2 = 27/8$, $\sigma_3 = 45/8$, $c_a = 5$)

three zones in the same way as $\mathbf{v}$ in our first-order model. Once the desired velocity has been determined, the velocity changes toward $\mathbf{v}_d$. We use

$$\mathbf{v}_d = H_{\sigma_1} \star \rho + f(\rho)\big(\big(G_{\sigma_2} \star (\rho\mathbf{v})\big)(G_{\sigma_2} \star \rho)^{-1} + c_a K_{\sigma_3} \star \rho\big), \tag{25}$$

$$\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} = \kappa(\mathbf{v}_d - \mathbf{v}), \tag{26}$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \tag{27}$$

We perturb both the density and velocity of an infinite uniform swarm by plane waves, taking $\rho^{(1)} = Ae^{i(\xi \cdot \mathbf{x} - \omega t)}$ and $\mathbf{v}^{(1)} = \mathbf{B}e^{i(\xi \cdot \mathbf{x} - \omega t)}$. When the velocity is constant, we have $\mathbf{v} = \mathbf{v}_d$, and so we use $\mathbf{v} = \mathbf{v}_0 + \epsilon \mathbf{v}^{(1)}$ and $\mathbf{v}_d = \mathbf{v}_0 + \epsilon \mathbf{v}_d^{(1)}$.

Again, we assume that the density is low enough that $f(\rho_0) = 1$ and $f'(\rho_0) = 0$. Then

$$\frac{\partial \mathbf{v}^{(1)}}{\partial t} + \mathbf{v}_0 \cdot \nabla \mathbf{v}^{(1)} = \kappa\big(H_{\sigma_1} \star \rho^{(1)} + G_{\sigma_2} \star \mathbf{v}^{(1)} + c_a K_{\sigma_3} \star \rho^{(1)} - \mathbf{v}^{(1)}\big). \tag{28}$$

Similar to the previous stability analysis, we take the Fourier transform of (28) and solve for $\widehat{\mathbf{v}}^{(1)}$. Applying the continuity equation gives us

$$\omega\widehat{\rho}^{(1)} + \mathbf{v}_0 \cdot \xi \widehat{\rho}^{(1)} + \rho_0 \kappa \widehat{\rho}^{(1)} \xi \cdot \big(\widehat{H}_{\sigma_1} + c_a \widehat{K}_{\sigma_3}\big)\big(-i\omega - i\mathbf{v}_0 \cdot \xi + \kappa - \kappa \widehat{G}_{\sigma_2}\big)^{-1} = 0. \tag{29}$$

Solving for ω gives us the dispersion relation

$$\omega_\pm = -\mathbf{v}_0 \cdot \xi - \frac{i\kappa}{2}\big(1 - e^{-\sigma_2^2|\xi|^2}\big) \pm i\sqrt{\left(\frac{\kappa}{2}\big(1 - e^{-\sigma_2^2|\xi|^2}\big)\right)^2 - 4\kappa\rho_0|\xi|^2\left(e^{-\sigma_1^2|\xi|^2} + c_a\left(\frac{\sigma_3^2}{2}|\xi|^2 - 1\right)e^{-\sigma_3^2|\xi|^2}\right)}. \tag{30}$$

See Appendix B for a more detailed treatment of the linear stability analysis.

Notice that if the discriminant is negative, then $\Im(\omega_\pm) = -\frac{\kappa}{2}(1 - e^{-\sigma_2^2|\xi|^2})$ is always negative. If the discriminant is positive, we have $\Im(\omega_-) < 0$ also. The imaginary part of ω_+ will be positive when

$$\frac{\kappa}{2}\left(1 - e^{-\sigma_2^2|\xi|^2}\right) < \sqrt{\left(\frac{\kappa}{2}\left(1 - e^{-\sigma_2^2|\xi|^2}\right)\right)^2 - 4\kappa\rho_0|\xi|^2\left(e^{-\sigma_1^2|\xi|^2} + c_a\left(\frac{\sigma_3^2}{2}|\xi|^2 - 1\right)e^{-\sigma_3^2|\xi|^2}\right)} \tag{31}$$

which is equivalent to

$$e^{-\sigma_1^2|\xi|^2} + c_a\left(\frac{\sigma_3^2}{2}|\xi|^2 - 1\right)e^{-\sigma_3^2|\xi|^2} < 0. \tag{32}$$

Note that in the first-order model, $\Im(\omega) > 0$ is equivalent to this same inequality. From our analysis in Sect. 4, we know that for $c_a > 1$ there is an interval of wavenumbers around $|\xi| = 0$ that makes the above inequality true. Again, the uniform swarm is stable unless $c_a > 1$ and $|\xi|$ lies in this interval.

For the individual-based simulations, we consider a large finite swarm on a regularly spaced lattice without periodic boundary conditions. Using Euler's method to numerically integrate (26) gives

$$\mathbf{v}_i(t+\tau) = \mathbf{v}_i(t) + \tau\kappa\big(\mathbf{v}_{d,i}(t) - \mathbf{v}_i(t)\big), \quad i = 1, \ldots, N. \tag{33}$$

Once the velocity vectors are determined, positions are integrated forwards in time as described in Sect. 3. Simulations of the individual-based model were performed on swarms of size $N = 2401$ with $\sigma_1 = 3/4$, $\sigma_2 = 9/4$, $\sigma_3 = 15/4$, $\kappa = 1$, and $\tau = 0.1$ and verified using a higher-order solver in SciPy. Figure 8 shows the results of the simulations with $c_a = 0.5$ and $c_a = 5$. The individual-based simulations verify the linear stability analysis and indicate the presence of a boundary layer for approximately $22 < r < 25$. When $c_a = 0.5$, outside of the boundary layer, $A(r,t)$ is nearly zero for short times so the swarm does not change configuration as predicted by the linear stability analysis. In contrast, when $c_a = 5$ the graph of $A(r,t)$ shows the swarm changes configuration and so there is a clear instability. Increasing the standard deviations of the behavioral interaction kernels increases the size of the boundary layer as expected. Just as in the first-order kinematic model, in the unstable parameter regime, we note the presence of an attractor corresponding to a circular, translating swarm that does not have a constant density.

6 The Second-Order (Dynamic), Constant-Speed Model

The second-order constant-speed model is close to the second-order variable speed model *and* close to the CKJRF constant-speed second-order discrete-time

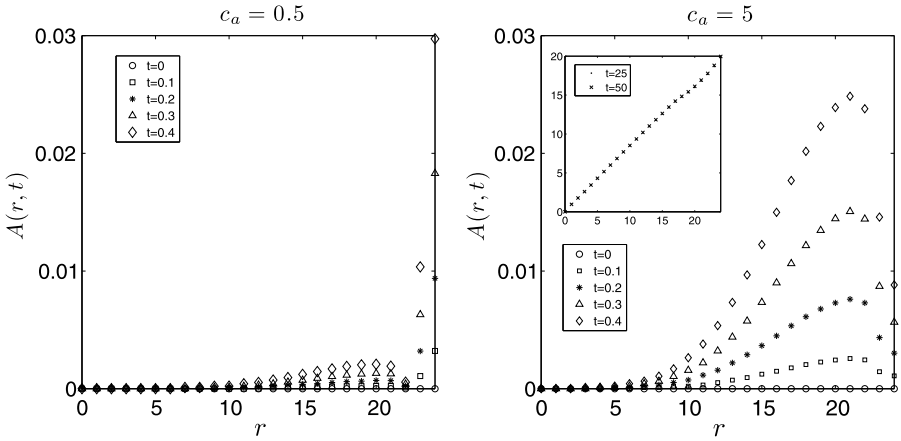


Fig. 8 Comparison of the second-order variable-speed individual-based model dynamics when $c_a = 0.5$ (stable) and $c_a = 5$ (unstable). The results are for a swarm of size $N = 2401$ with $\sigma_1 = 3/4$, $\sigma_2 = 9/4$, and $\sigma_3 = 15/4$. A coarse-observable $A(r,t)$ is used to measure the magnitude of change in configuration over time as a function of distance from the interior

model (Couzin et al. 2002). A continuous-time dynamical systems model corresponding to Couzin et al. (2002) was proposed in Paley et al. (2007a) and analyzed using control theoretic techniques. Here, we develop a continuum version of this model.

Our second-order constant-speed model uses the density ρ and the angle of direction θ to describe the swarm. The unit velocity $\mathbf{v}$ is determined by the angle since the speed is constant, and $\mathbf{v}_d$ is defined in the same way as the previous models. The material derivative of θ is proportional to the length of the projection of $\mathbf{v}_d$ onto $\mathbf{v}^\perp$.

$$\mathbf{v} = \langle \cos\theta, \sin\theta \rangle, \tag{34}$$

$$\mathbf{v}^\perp = \langle -\sin\theta, \cos\theta \rangle, \tag{35}$$

$$\mathbf{v}_d = H_{\sigma_1} \star \rho + f(\rho)\big(\big(G_{\sigma_2} \star (\rho\mathbf{v})\big)(G_{\sigma_2} \star \rho)^{-1} + c_a K_{\sigma_3} \star \rho\big), \tag{36}$$

$$\frac{\partial\theta}{\partial t} + (\mathbf{v}\cdot\nabla)\theta = \kappa\mathbf{v}^\perp \cdot \mathbf{v}_d, \tag{37}$$

$$\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\mathbf{v}) = 0. \tag{38}$$

We perturb θ and ρ in a uniform swarm by letting $\theta = \theta_0 + \epsilon\theta^{(1)}$ and $\rho = \rho_0 + \epsilon\rho^{(1)}$, where θ_0 and ρ_0 are constant. Expanding $\mathbf{v}$ and $\mathbf{v}^\perp$ around $\epsilon = 0$, we obtain $\mathbf{v} = \mathbf{v}_0 + \epsilon\theta^{(1)}\mathbf{v}_0^\perp$ and $\mathbf{v}^\perp = \mathbf{v}_0^\perp - \epsilon\theta^{(1)}\mathbf{v}_0$, where $\mathbf{v}_0 = \langle \cos\theta_0, \sin\theta_0 \rangle$ and $\mathbf{v}_0^\perp = \langle -\sin\theta_0, \cos\theta_0 \rangle$. Then we have

$$\begin{aligned}\frac{\partial\theta^{(1)}}{\partial t} &+ \mathbf{v}_0 \cdot \nabla\theta^{(1)} \\ &= \kappa\big(\mathbf{v}_0^\perp \cdot H_{\sigma_1} \star \rho^{(1)} + f(\rho_0)\big(G_{\sigma_2} \star \theta^{(1)} + c_a\mathbf{v}_0^\perp \cdot K_{\sigma_3} \star \rho^{(1)} - \theta^{(1)}\big)\big)\end{aligned} \tag{39}$$

assuming that $\|\mathbf{v}_0\| = \|\mathbf{v}_0^\perp\| = 1$. See Appendix C for details of the expansion.

Now assume that $f(\rho_0) = 1$ and let $\rho^{(1)} = A_1 e^{i(\xi\cdot\mathbf{x}-\omega t)}$, $\theta^{(1)} = A_2 e^{i(\xi\cdot\mathbf{x}-\omega t)}$. Making these substitutions in (39) and the conservation equation leads us to the system

$$\begin{bmatrix} \mathbf{v}_0\cdot\xi+\omega & \rho_0\mathbf{v}_0^{\perp}\cdot\xi \\ \kappa\mathbf{v}_0^{\perp}\cdot(\widehat{H}_{\sigma_1}+c_a\widehat{K}_{\sigma_3}) & -\kappa(1-\widehat{G}_{\sigma_2})+i(\mathbf{v}_0\cdot\xi+\omega) \end{bmatrix}\begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \tag{40}$$

The determinant of the matrix must be zero in order to have nontrivial solutions to (40). We solve for ω to find the dispersion relation

$$\begin{aligned}\omega_{\pm} &= -\mathbf{v}_0\cdot\xi - \frac{i\kappa}{2}\left(1-e^{-\sigma_2^2|\xi|^2}\right) \\ &\quad \pm i\sqrt{\left(\frac{\kappa}{2}\left(1-e^{-\sigma_2^2|\xi|^2}\right)\right)^2 - 4\kappa\rho_0|\xi|^2\left(e^{-\sigma_1^2|\xi|^2}+c_a\left(\frac{\sigma_3^2}{2}|\xi|^2-1\right)e^{-\sigma_3^2|\xi|^2}\right)}.\end{aligned} \tag{41}$$

This is the same dispersion relation as for the second-order variable-speed model in Sect. 5. Once again, there is an interval of wavenumbers around $|\xi| = 0$ such that perturbations with these wavenumbers will grow exponentially if $c_a > 1$. Otherwise, the swarm is stable.

As before, for the individual-based simulations, we simulate a large finite swarm on a regularly spaced lattice without imposing periodic boundary conditions. Using Euler's method to numerically integrate (36) gives

$$\theta_i(t+\tau) = \theta_i(t) + \tau\kappa\mathbf{v}_i(t)^{\perp}\cdot\mathbf{v}_{d,i}(t), \quad i = 1,\ldots,N, \tag{42}$$

where the velocity vector is defined as $\mathbf{v}_i(t) = \langle\cos(\theta_i(t)), \sin(\theta_i(t))\rangle$. Once the velocity vectors are determined, positions are integrated forward in time as described in Sect. 3.

The stability results of individual-based simulations of the second-order constant-speed model correspond to those of the second-order variable-speed model. Specifically, simulations of the individual-based model indicate the presence of a boundary layer. For sufficiently short times, outside of the boundary layer, the stability of the swarm (as characterized by the coarse observable $A(r,t)$) is determined by c_a. For long times, boundary effects become important. In stable parameter regimes, the swarm expands anisotropically. In unstable parameter regimes, the swarm evolves away from the uniform translating state toward a variable-density coherent anisotropic attractor. See Fig. 9 for an example when $c_a = 0.5$ and $c_a = 5$. In the unstable parameter regime, for c_a sufficiently large, individual-based simulations indicate that swarms can fragment as shown in Fig. 10.

We also simulate the full nonlinear model which includes the density dependent switching function $f(\rho)$ deactivating orientation and attraction when the population reaches a critical density. We implement this in the individual-based simulations by defining a critical approach distance r. If an individual has any neighbors within this approach distance, the behavioral input vector $\mathbf{v}_{d,i} = \mathbf{v}_{r,i}$ so that there is no contribution from orientation or attraction. Such an approach is used in Couzin et al. (2002) where the critical approach distance r is defined to be the radius of the zone of repulsion. Our steady-state simulations indicate the presence of an incoherent swarming

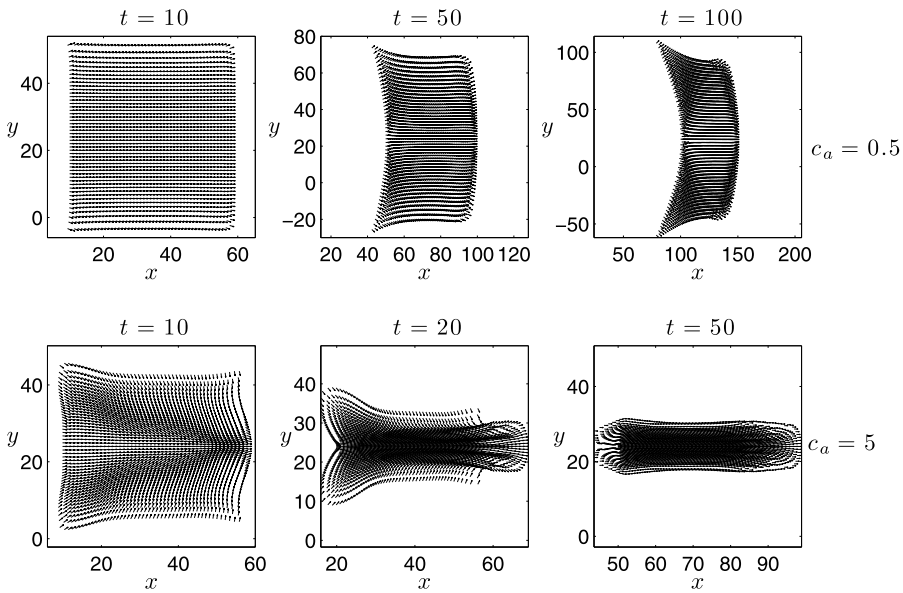


Fig. 9 Evolution of a constant-density traveling swarm on a square lattice simulated using the constant-speed second-order individual-based model with $N = 2401$, $\sigma_1 = 3/4$, $\sigma_2 = 9/4$, $\sigma_3 = 15/4$. When $c_a = 0.5$ (*top*), the uniform translating state expands anisotropically. When $c_a = 5$ (*bottom*), the swarm evolves away from the uniform translating state toward a variable-density attractor

state in parameter regimes for which constant-density traveling groups are unstable and a uniform translating state in stable parameter regimes. Figure 11 shows an example when $r = 0.75$, $\kappa = 10$, and $c_a = 100$ and $c_a = 1$, respectively.

The incoherent and translating state are often present as steady-states in individual-based models; see, e.g., Couzin et al. (2002), Li et al. (2008), Vicsek et al. (1995). Li et al. (2008) considered a two-dimensional individual-based model of swarming similar to our own and noted a shift in stability from the incoherent swarming state to a highly parallel translating state as the proportion of attraction to alignment tendencies changed. This parameter plays a similar role to the parameter c_a in our model.

7 Conclusions and Future Work

In this paper, we have presented three new three-zone continuum models that range from a simple first-order model to a constant-speed second-order model that is aligned with popular individual-based models, particularly those in Couzin et al. (2002) and Paley et al. (2007b). For each of these models, an infinite uniform swarm is a stationary solution and we have analyzed its linear stability. The weight c_a of the attraction term determines whether the swarm is stable. When $c_a > 1$, there is an interval of wavenumbers near zero that will cause instability in the swarm. The zone sizes for repulsion and attraction, σ_1 and σ_3, fix this interval of wavenumbers. We note that in discrete systems, the smallest possible wavenumber might be outside of

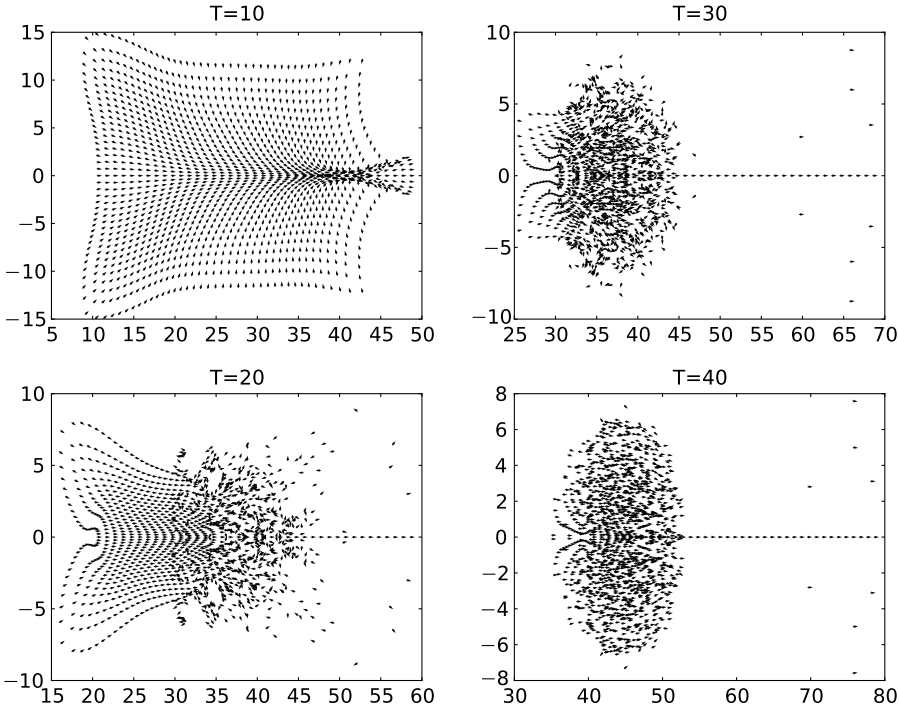


Fig. 10 Evolution of constant-density swarm on a square lattice with the same parameters as in Fig. 9 but with $c_a = 10$. The strong attractive force from individuals on the boundary causes the swarm to collapse and fragment

this interval. Therefore, in small aggregations of individuals, the system can be stable even though $c_a > 1$.

In order to assess the validity of these predictions, we have compared the results of our analysis to the results from individual-based models using the same interaction rules. In our individual-based model simulations, we observe a boundary layer. The swarms used are finite but large, and the central regions of the swarms are analogous to regions of infinite uniform swarms. Because the first-order model has individuals interact and react instantaneously, the effects of the boundary are felt immediately throughout the swarm. In the second-order models, the interior of the swarm does not feel the effects of the boundary layer for short timescales, so that there is a period of time when the behavior in the interior is indicative of the behavior of an infinite uniform swarm. In this regime, we have verified our theoretical results using the individual-based model. In all cases, $c_a < 1$ will lead to stable translating uniform swarms as predicted. For the first-order model, we observe that when $c_a > 1$, the swarm will evolve toward a circular attractor with a variable density distribution. We were able to reproduce this solution by numerically solving the integral equation corresponding to an axisymmetric state balancing attraction and repulsion. An analytic description of this circular attractor is a subject for future research. In the unstable regime, the second-order models cause aggregations to evolve toward more complex attractors and sometimes fragment.

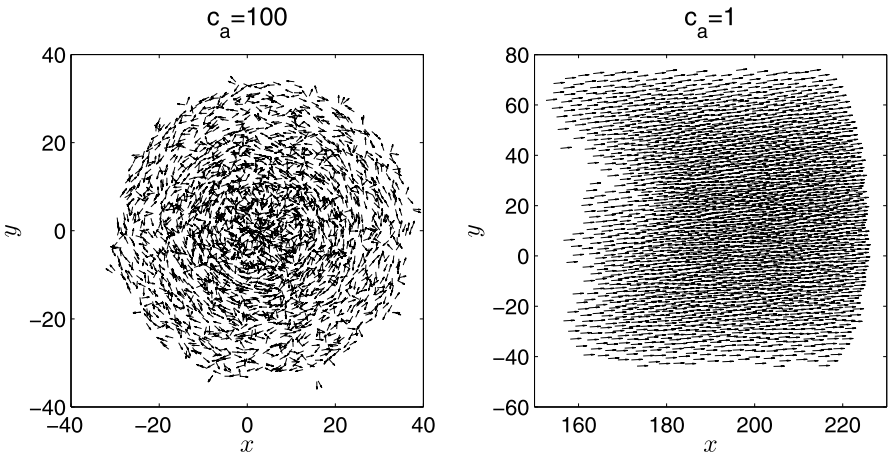


Fig. 11 Individual-based simulations of the full nonlinear model at steady-state reveal an incoherent swarming state and a translating state as the parameter c_a is varied. The results are for a swarm of size $N = 2401$ with $\sigma_1 = 3/4$, $\sigma_2 = 9/4$, and $\sigma_3 = 15/4$, $r = 0.75$, and $\kappa = 10$

There are directions along which one can extend this model and the techniques employed in this paper. We are extending these models to include leadership, wherein one or more individuals have additional information. This extra information could be an intended destination, for example, which means that including leadership may capture processes such as migration. In the case of the first-order model, including leadership would also resolve the nontrivial null space issue for at least part of the swarm since leaders would have a preferred direction even without input from social interactions. Another extension is to use anisotropic social interaction kernels. For example, elliptic zones rather than circular could simulate an individual who is able to receive more information from some directions. Finally, further investigation into the observed variable density attractor may lead to insights into the properties of other useful coherent structures, such as milling, that are observed in popular individual-based models and natural systems.

Acknowledgements LFR and JMM acknowledge the support of National Science Foundation Grants CCF-0726556 and CCF-0829748 for supporting this work. JPJN acknowledges the support of the UD Mathematical Sciences GEMS program. AK acknowledges the support of the Unidel Foundation.

Appendix A: Linear Stability Analysis of the First-Order Model

Suppose that $\rho(\mathbf{x}, t) = \rho_0 + \epsilon\rho^{(1)}(\mathbf{x}, t)$ and $\mathbf{v}(\mathbf{x}, t) = \mathbf{v}_0 + \epsilon\mathbf{v}^{(1)}(\mathbf{x}, t)$. Expanding around $\epsilon = 0$ and collecting terms of order ϵ, the behavior equation (12) gives us

$$\mathbf{v}^{(1)} = H_{\sigma_1} \star \rho^{(1)} + f(\rho_0)\big(G_{\sigma_2} \star \mathbf{v}^{(1)} + c_a K_{\sigma_3} \star \rho^{(1)}\big) + f'(\rho_0)\mathbf{v}_0\rho^{(1)} \tag{43}$$

and the continuity equation (11) becomes

$$\frac{\partial\rho^{(1)}}{\partial t} + \mathbf{v}_0 \cdot \nabla\rho^{(1)} + \rho_0\nabla \cdot \mathbf{v}^{(1)} = 0. \tag{44}$$

Assuming that $f(\rho_0) = 1$ and $f'(\rho_0) = 0$, our equation for $\mathbf{v}^{(1)}$ becomes

$$\mathbf{v}^{(1)} = H_{\sigma_1} \star \rho^{(1)} + G_{\sigma_2} \star \mathbf{v}^{(1)} + c_a K_{\sigma_3} \star \rho^{(1)}. \tag{45}$$

We use a plane wave $\rho^{(1)} = Ae^{i(\xi\cdot\mathbf{x}-\omega t)}$ as the perturbation in the density and take the Fourier transform $\widehat{g}(\xi) = \int_{-\infty}^{\infty} g(x)e^{i\xi x}\,dx$. Then we can solve for $\widehat{\mathbf{v}}^{(1)}$, and we have

$$\begin{aligned}\widehat{\mathbf{v}}^{(1)} &= \frac{(\widehat{H}_{\sigma_1} + c_a\widehat{K}_{\sigma_3})\widehat{\rho}^{(1)}}{(1-\widehat{G}_{\sigma_2})}\\ &= \frac{i\xi(e^{-\sigma_1^2|\xi|^2} + c_a(\frac{\sigma_3^2}{2}|\xi|^2 - 1)e^{-\sigma_3^2|\xi|^2})\widehat{\rho}^{(1)}}{1-e^{-\sigma_2^2|\xi|^2}}.\end{aligned} \tag{46}$$

The Fourier transform of (44) is $i\omega\widehat{\rho}^{(1)} + \mathbf{v}_0\cdot(i\xi\widehat{\rho}^{(1)}) + i\rho_0\xi\cdot\widehat{\mathbf{v}}^{(1)} = 0$. Replacing $\widehat{\mathbf{v}}^{(1)}$ with (46) and solving for ω, we obtain the dispersion relation

$$\omega = -\mathbf{v}_0\cdot\xi - i\rho_0|\xi|^2\frac{e^{-\sigma_1^2|\xi|^2} + c_a(\frac{\sigma_3^2}{2}|\xi|^2-1)e^{-\sigma_3^2|\xi|^2}}{1-e^{-\sigma_2^2|\xi|^2}}. \tag{47}$$

Appendix B: Linear Stability Analysis of the Second-Order, Variable-Speed Model

The order ϵ perturbation in the desired velocity is

$$\mathbf{v}_d^{(1)} = H_{\sigma_1}\star\rho^{(1)} + f(\rho_0)\big(G_{\sigma_2}\star\mathbf{v}^{(1)} + c_a K_{\sigma_3}\star\rho^{(1)}\big) + f'(\rho_0)\rho^{(1)}\mathbf{v}_0. \tag{48}$$

From (26) and (27) at order ϵ, we obtain

$$\frac{\partial\mathbf{v}^{(1)}}{\partial t} + \mathbf{v}_0\cdot\nabla\mathbf{v}^{(1)} = \kappa\big(\mathbf{v}_d^{(1)} - \mathbf{v}^{(1)}\big), \tag{49}$$

$$\frac{\partial\rho^{(1)}}{\partial t} + \mathbf{v}_0\cdot\nabla\rho^{(1)} + \rho_0\nabla\cdot\mathbf{v}^{(1)} = 0. \tag{50}$$

Again, we assume that the density is low enough that $f(\rho_0) = 1$ and $f'(\rho_0) = 0$. Substituting (48) into (49),

$$\frac{\partial\mathbf{v}^{(1)}}{\partial t} + \mathbf{v}_0\cdot\nabla\mathbf{v}^{(1)} = \kappa\big(H_{\sigma_1}\star\rho^{(1)} + G_{\sigma_2}\star\mathbf{v}^{(1)} + c_a K_{\sigma_3}\star\rho^{(1)} - \mathbf{v}^{(1)}\big). \tag{51}$$

We use a plane wave to perturb ρ and $\mathbf{v}$, so that $\rho^{(1)} = Ae^{i(\xi\cdot\mathbf{x}-\omega t)}$, $\mathbf{v}^{(1)} = \mathbf{B}e^{i(\xi\cdot\mathbf{x}-\omega t)}$. Take the Fourier transform of (51) and then solve for $\widehat{\mathbf{v}}^{(1)}$ to obtain

$$\widehat{\mathbf{v}}^{(1)} = \kappa\widehat{\rho}^{(1)}\big(\widehat{H}_{\sigma_1} + c_a\widehat{K}_{\sigma_3}\big)\big(-i\omega - i\mathbf{v}_0\cdot\xi + \kappa - \kappa\widehat{G}_{\sigma_2}\big)^{-1}. \tag{52}$$

The Fourier transform of (50) is $\omega\widehat{\rho}^{(1)} + \mathbf{v}_0 \cdot \xi\widehat{\rho}^{(1)} + \rho_0\widehat{\mathbf{v}}^{(1)} \cdot \xi = 0$. Substituting for $\widehat{\mathbf{v}}^{(1)}$ in this equation gives us

$$\omega\widehat{\rho}^{(1)} + \mathbf{v}_0 \cdot \xi\widehat{\rho}^{(1)} + \rho_0\kappa\widehat{\rho}^{(1)}\xi \cdot \big(\widehat{H}_{\sigma_1} + c_a\widehat{K}_{\sigma_3}\big)\big(-i\omega - i\mathbf{v}_0 \cdot \xi + \kappa - \kappa\widehat{G}_{\sigma_2}\big)^{-1} = 0. \tag{53}$$

Solving for ω gives us the dispersion relation

$$\omega_\pm = -\mathbf{v}_0 \cdot \xi - \frac{i\kappa}{2}\big(1 - e^{-\sigma_2^2|\xi|^2}\big) \pm i\sqrt{\left(\frac{\kappa}{2}\big(1 - e^{-\sigma_2^2|\xi|^2}\big)\right)^2 - 4\kappa\rho_0|\xi|^2\left(e^{-\sigma_1^2|\xi|^2} + c_a\left(\frac{\sigma_3^2}{2}|\xi|^2 - 1\right)e^{-\sigma_3^2|\xi|^2}\right)}. \tag{54}$$

Appendix C: Linear Stability Analysis of the Second-Order, Constant-Speed Model

As before, we perturb θ and ρ in a uniform swarm by letting $\theta = \theta_0 + \epsilon\theta^{(1)}$ and $\rho = \rho_0 + \epsilon\rho^{(1)}$, where θ_0 and ρ_0 are constant.

Expanding $\mathbf{v}$ and $\mathbf{v}^\perp$ around $\epsilon = 0$, we have

$$\mathbf{v} = \langle\cos\theta_0, \sin\theta_0\rangle + \epsilon\theta^{(1)}\langle-\sin\theta_0, \cos\theta_0\rangle, \tag{55}$$

$$\mathbf{v}^\perp = \langle-\sin\theta_0, \cos\theta_0\rangle - \epsilon\theta^{(1)}\langle\cos\theta_0, \sin\theta_0\rangle. \tag{56}$$

Define $\mathbf{v}_0 = \langle\cos\theta_0, \sin\theta_0\rangle$, $\mathbf{v}_0^\perp = \langle-\sin\theta_0, \cos\theta_0\rangle$, so that $\mathbf{v} = \mathbf{v}_0 + \epsilon\theta^{(1)}\mathbf{v}_0^\perp$ and $\mathbf{v}^\perp = \mathbf{v}_0^\perp - \epsilon\theta^{(1)}\mathbf{v}_0$.

We expand (36) around $\epsilon = 0$, and at order ϵ we have

$$\begin{aligned}
&\frac{\partial\theta^{(1)}}{\partial t} + \mathbf{v}_0 \cdot \nabla\theta^{(1)} \\
&\quad = \kappa\mathbf{v}_0^\perp \cdot \Bigg[H_{\sigma_1} \star \rho^{(1)} + f(\rho_0)\rho_0\mathbf{v}_0\left(\frac{1}{\rho_0^2}G_{\sigma_2} \star \rho^{(1)}\right) \\
&\qquad + \frac{1}{\rho_0}f(\rho_0)\big(\rho_0\mathbf{v}_0^\perp G_{\sigma_2} \star \theta^{(1)} + v_0 G_{\sigma_2} \star \rho^{(1)}\big) + f(\rho_0)c_a K_{\sigma_3} \star \rho^{(1)} \\
&\qquad + \rho^{(1)}f'(\rho_0)(\rho_0\mathbf{v}_0)\left(\frac{1}{\rho_0}\right)\Bigg] - \kappa\theta^{(1)}\mathbf{v}_0 \cdot (f(\rho_0)\rho_0\mathbf{v}_0)\left(\frac{1}{\rho_0}\right)
\end{aligned} \tag{57}$$

which simplifies to

$$\frac{\partial\theta^{(1)}}{\partial t} + \mathbf{v}_0 \cdot \nabla\theta^{(1)} = \kappa\big(\mathbf{v}_0^\perp \cdot H_{\sigma_1} \star \rho^{(1)} + f(\rho_0)\big(G_{\sigma_2} \star \theta^{(1)} + c_a\mathbf{v}_0^\perp \cdot K_{\sigma_3} \star \rho^{(1)} - \theta^{(1)}\big)\big) \tag{58}$$

assuming that $\|\mathbf{v}_0\| = \|\mathbf{v}_0^\perp\| = 1$. From the continuity equation at $O(\epsilon)$, we have $\frac{\partial\rho^{(1)}}{\partial t} + \mathbf{v}_0 \cdot \nabla\rho^{(1)} + \rho_0\mathbf{v}_0^\perp \cdot \nabla\theta^{(1)} = 0$.

Now assume that $f(\rho_0) = 1$ and let $\rho^{(1)} = A_1 e^{i(\xi\cdot\mathbf{x}-\omega t)}$, $\theta^{(1)} = A_2 e^{i(\xi\cdot\mathbf{x}-\omega t)}$. Taking the Fourier transforms of the equation for $\frac{\partial\theta^{(1)}}{\partial t}$ and the continuity equation gives us

$$\frac{\partial\widehat{\theta}^{(1)}}{\partial t} - i\xi\cdot\mathbf{v}_0\theta^{(1)} = \kappa\big(\mathbf{v}_0^{\perp}\cdot\widehat{H}_{\sigma_1}\widehat{\rho}^{(1)} + \widehat{G}_{\sigma_2}\widehat{\theta}^{(1)} + c_a\mathbf{v}_0^{\perp}\cdot\widehat{K}_{\sigma_3}\widehat{\rho}^{(1)} - \widehat{\theta}^{(1)}\big), \tag{59}$$

$$\frac{\partial\widehat{\rho}^{(1)}}{\partial t} - i\xi\cdot\mathbf{v}_0\widehat{\rho}^{(1)} - i\rho_0\xi\cdot\mathbf{v}_0^{\perp}\widehat{\theta}^{(1)} = 0. \tag{60}$$

Substituting for $\widehat{\theta}^{(1)}$ and $\widehat{\rho}^{(1)}$, the continuity equation gives us

$$-iA_1(\omega + \mathbf{v}_0\cdot\xi) - iA_2\rho_0\mathbf{v}_0^{\perp}\cdot\xi = 0. \tag{61}$$

Substituting for $\widehat{\theta}^{(1)}$ and $\widehat{\rho}^{(1)}$ in (59) gives us

$$-iA_2\omega - iA_2\mathbf{v}_0\cdot\xi = \kappa\big(A_1\mathbf{v}_0^{\perp}\cdot\widehat{H}_{\sigma_1} + A_2\widehat{G}_{\sigma_2} + A_1c_a\mathbf{v}_0^{\perp}\cdot\widehat{K}_{\sigma_3} - A_2\big) \tag{62}$$

so that

$$A_1\kappa\big(\mathbf{v}_0^{\perp}\cdot\widehat{H}_{\sigma_1} + c_a\mathbf{v}_0^{\perp}\cdot\widehat{K}_{\sigma_3}\big) + A_2\big(\kappa\widehat{G}_{\sigma_2} - \kappa + i\omega + i\mathbf{v}_0\cdot\xi\big) = 0. \tag{63}$$

The system that must be satisfied is

$$\begin{bmatrix} \mathbf{v}_0\cdot\xi + \omega & \rho_0\mathbf{v}_0^{\perp}\cdot\xi \\ \kappa\mathbf{v}_0^{\perp}\cdot(\widehat{H}_{\sigma_1} + c_a\widehat{K}_{\sigma_3}) & -\kappa(1-\widehat{G}_{\sigma_2}) + i(\mathbf{v}_0\cdot\xi + \omega) \end{bmatrix}\begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \tag{64}$$

References

Aoki, I. (1982). A simulation study on the schooling mechanism in fish. *Bull. Jpn. Soc. Sci. Fisher.*, *48*, 1081–1088.

Bonabeau, E. (1999). Scaling in animal group-size distributions. *Proc. Natl. Acad. Sci. USA*, *96*(8), 4472–4477.

Bonabeau, E., & Dagorn, L. (1995). Possible universality in the size distribution of fish schools. *Phys. Rev. E*, *51*(6), R5220–R5223.

Bonnans, J.-F., Gilbert, J. C., Lemarechal, C., & Sagatizabal, C. A. (2006). *Numerical optimization—Theoretical and Practical Aspects*. New York: Springer.

Camazine, S., Deneubourg, J.-L., Franks, N. R., Sneyd, J., Theraula, G., & Bonabeau, E. (2003). *Self-organization in biological systems*. Princeton: Princeton University Press.

Couzin, I., Krause, J., James, R., Ruxton, G., & Franks, N. (2002). Collective memory and spatial sorting in animal groups. *J. Theor. Biol.*, *218*, 1–11.

Cucker, F., & Smale, S. (2007). Emergent behavior in flocks. *IEEE Trans. Autom. Control*, *52*(5), 852–862.

Eftimie, R., de Vries, G., Lewis, M.a., & Lutscher, F. (2007). Modeling group formation and activity patterns in self-organizing collectives of individuals. *Bull. Math. Biol.*, *69*(5), 1537–1565.

Giardina, I. (2008). Collective behavior in animal groups: theoretical models and empirical studies. *HFSP J.*, *2*(4), 205–219.

Huepe, C., & Aldana, M. (2004). Intermittency and clustering in a system of self-driven particles. *Phys. Rev. Lett.*, *92*, 168701.

Huth, A., & Wissel, C. (1992). The simulation of the movement of fish schools. *J. Theor. Biol.*, *156*, 365–385.

Kolpas, A., & Moehlis, J. (2009). Optimal switching between collective motion states for two agents. *Appl. Math. Lett.*, *22*(4), 4.

Kolpas, A., Moehlis, J., & Kevrekidis, I. G. (2007). Coarse-grained analysis of stochasticity-induced switching between collective motion states. *Proc. Natl. Acad. Sci. USA*, *104*(14), 5931–5935.

Levine, H., Rappel, W. J., & Cohen, I. (2001). Self-organization in systems of self-propelled particles. *Phys. Rev. E*, *63*, 017101.

Li, H., Kolpas, A., Moehlis, J., & Petzold, L. (2008). Parallel simulation for a fish schooling model on a general-purpose graphics processing unit. *Concurr. Comput. Pract. Exp.* doi:10.1002/cpe.1330.

Lissaman, P. B. S., & Shollenberger, C. A. (1970). Formation flight of birds. *Science*, *168*, 1003–1005.

Lukeman, R., Li, Y.-X., & Edelstein-Keshet, L. (2010). Inferring individual rules from collective behavior. *Proc. Natl. Acad. Sci. USA*, *107*(28), 12576–12580.

Mogilner, A., & Edelstein-Keshet, L. (1999). A non-local model for a swarm. *J. Math. Biol.*, *38*(6), 534–570.

Niwa, H.-S. (2003). Power-law versus exponential distributions of animal group sizes. *J. Theor. Biol.*, *224*(4), 451–457.

Paley, D., Leonard, N., Sepulchre, R., Grünbaum, D., & Parrish, J. K. (2007a). Spatial patterns in the dynamics of engineered and biological networks. *IEEE Control Syst. Mag.*, *27*(4), 89–105.

Paley, D. A., Leonard, N. E., Sepulchre, R., Grübaum, D., & Parrish, J. K. (2007b). Oscillator models and collective motion. *IEEE Control Syst. Mag.*, 89–105. doi:10.1109/MCS.2007.384123.

Parrish, J. K. (1999). Complexity, pattern, and evolutionary trade-offs in animal aggregation. *Science*, *284*(5411), 99–101.

Parrish, J. K., Viscido, S. V., & Grünbaum, D. (2002). Self-organized fish schools: an examination of emergent properties. *Biol. Bull.*, *202*, 296–305.

Partridge, B. (1982). The structure and function of fish schools. *Sci. Am.* doi:10.1038/scientificamerican0682-114.

Penrose, R., & Todd, J. A. (1956). On best approximate solutions of linear matrix equations. *Math. Proc. Camb. Philos. Soc.*, *52*(01), 17–19.

Raymond, J. R., & Evans, M. R. (2006). Flocking regimes in a simple lattice model. *Phys. Rev. E*, *73*, 36112.

Reuter, H., & Breckling, B. (1994). Self-organization of fish schools: an object-oriented model. *Ecol. Model.*, *75*, 147–159.

Sinclair, A. R. E., & Norton-Griffiths, M. (1979). *Serengeti: dynamics of an ecosystem*. Chicago: University of Chicago.

Topaz, C., & Bertozzi, A. (2004). Swarming patterns in a two-dimensional kinematic model for biological groups. *SIAM J. Appl. Math.* doi:10.1137/S00361399034377424.

Topaz, C. M., Bertozzi, A. L., & Lewis, M. A. (2006). A nonlocal continuum model for biological aggregation. *Bull. Math. Biol.*, *68*(7), 1601–1623.

Uvarov, B. P. (1928). *Grasshoppers and locusts*. London: Imperial Bureau of Entomology.

Vicsek, T., Czirok, A., Ben-Jacob, E., Cohen, I., & Shochet, O. (1995). Novel type of phase transition in a system of self-driven particles. *Phys. Rev. Lett.*, *75*, 1226–1229.

Warburton, K., & Lazarus, J. (1991). Tendency-distance models of social cohesion in animal groups. *J. Theor. Biol.*, *150*, 473–488.